PM FELLOWSHIP CASE STUDY · CORE EXPERIENCE

Google Photos People remember the photo, but can’t find it

OBJECTIVE Raise the share of users who find a **half-remembered photo** in Google Photos, **quickly**.
Turn several fuzzy memory cues into the one photo they are thinking of.

LOOKING FOR AN OLD PHOTO · WHERE RECALL BREAKS

1 Recall Goa, Riya, Diwali → **2 Search** types just “Goa” → **3 Scan** scrolls hundreds → **4 Pick** is this the one? → **5 Give up** asks a friend instead

958M Indians online

1.5B Photos users a month

9T photos stored

370M searches a month

FINDING IS NOT SOLVED · POPSA 2026

70% of photos are never revisited

37% say important photos get lost

73% are open to AI help

HOW APPS “FIND” TODAY

Photos search	1 word
Ask Photos	1 sentence
WhatsApp	1 chat
Phone gallery	1 month

THE GAP IN ONE LINE

People remember several cues; search takes **one**, and no tool shows what it understood.

JOBS TO BE DONE

“Find Riya’s photo from Goa.”
“Get that bill photo, now.”
“Show the one I mean, not 400.”

SCOPE

IN Search and retrieval, India

FOR Photos you half remember

OUT Backup, editing, sharing

METHOD **1** Metric tree · **2** AI discovery · **3** Survey 46 → 6 interviews · **4** Problem → MoSCoW → MVP · **5** Metrics, guardrails, risks

Sources: Google Photos at 10 (2025) · IAMAI-Kantar 2025 · Popsa 2026 (n=8,000) · Ask Photos in India (2025)

Google Photos

What must change for a half-remembered photo to be found

NORTH STAR **Find rate ↑** Share of retrieval sessions where the wanted photo is opened or shared within 60 seconds

■ Describe the memory
Who, where, when + details inside the photo

■ Buried in results
Rank of photo opened · scroll depth · duplicates on top

☐ Not searchable
Photos never processed · face grouping off · outside library

☐ Misread by the model
Wrong dates or labels · exits after a wrong result

☐ Never backed up
Folders not backed up · storage full · backup off

WHERE TO PLAY

Describe the memory **5×4=20**

Photos reads text; AI reads details

Buried in results **4×4=16**

16% of phone photos are duplicates

Not searchable 4×2=8

Misread by the model 3×2=6

Never backed up 4×1=4

Score = **impact × ease**, 1–5 each

THE MOMENT WE DESIGN FOR

“me holding a black cup” · “man with a mole and freckles” · “that AC bill”

	Today	What we want
Describe	Types one word	Says it as they remember
Recognise	Scrolls hundreds	Few picks + why
Recover	One detail wrong: nothing	Close match still shows
Return	Starts from zero	Knows their people, places

It breaks at Describe: the box takes one word.

WHAT RIVAL APPS DO

4/4 take a sentence

0/4 show why it matched

0/4 fix a wrong cue

If they guess wrong, you can’t see why or fix it. Google paused Ask Photos in 2025 over quality.

Google, Apple, Samsung, OneDrive · public docs

That gap is our space

FIX

Describe the memory + buried

Top scores; the pieces already exist

Watch: 2+ detail searches ↑ · rank opened ↓

NOT NOW

Not searchable: never processed, faces off by region

Misread: wrong dates, labels

OUT

Never backed up: storage pricing

Sources: Diary study · arXiv 2021 · TechCrunch · Avast · 9to5Google · AOL · Apple · Samsung · Microsoft

Next: tested on real posts →

Google Photos — We read 54,000 public posts before asking users anything

5-STAGE PIPELINE · PUBLIC POST → SCORED INSIGHT

01 COLLECT — 7 public sites, Python scrapers → **02 CLEAN** — Cut repeats, keep search posts → **03 TAG** — Gemini fills 12 fields per post → **04 SCORE** — Rank each failure by opportunity → **05 PUBLISH** — Live site, insights, Ask tab

53,986	7	1,091	10
public posts read	sources	real search stories	insights · 0 hidden

STACK Python · Reddit, YouTube, HN, Play APIs · BeautifulSoup · Gemini flash-lite · Vercel

CORPUS FUNNEL · ONE RUN, OCT 2026

53,986 collected	100%	7 sites, since 2007
35,578 clean	66%	repeats, non-English
8,151 on topic	15%	can't find + photo
8,141 tagged	15%	10 unreadable
3,695 evidence	6.8%	storage, crash cut
1,091 stories	2%	real failed search

The other **2,604** evidence posts are feedback on search

DATA SOURCES · POSTS / STORIES

	Posts	Stories
Play Store	33,436	341
YouTube	11,196	51
Reddit	5,430	389
Help forum	2,106	202
Hacker News	1,211	74
Stack Exch.	418	25
App Store	189	9

Play Store gave volume. **Reddit gave stories.**

ANATOMY OF A PUBLISHED INSIGHT

ASK	What do people recall?
ANSWER	**A rough when, plus who or what.**
COUNTER	People write less than they recall.
ACTION	Treat time as a range, not a date.
PROOF	High 1,091 posts · 7 sites

TRUST LAYER

86% relevance
94% stage

80 posts re-read by a 2nd AI model. Every number opens its posts. Skews Android.

OPPORTUNITY SCORE

`√posts×(.5+pain)×(.5+spread)×fix`

common · painful · on many sites · fixable

Can't combine cues	30.4
Buried in results	30.2
Not searchable	29.9
Misreads the query	28.5
Search got worse	19.0

Top four within 2 points: one tier

HOW IT FEEDS RESEARCH

Public posts → Engine → 6 hunches → Survey + interviews → **Core problem**

Engine says **what to investigate**. Users say **what is true**.

Live engine · Code on GitHub

Google Photos

Ten questions answered: search failing the memory, not lost photos, dominates the evidence

INSIGHTS BY QUESTION · ANSWER		POSTS · EVIDENCE	
What they remember	rough time + who or what	1,091	High
What they type	one or two words	235	Med
Posts back our focus	combining cues, buried	1,814	High
Why combining fails	the second cue is dropped	452	High
Why results bury it	volume, wrong dates	347	High
What they do instead	scroll or give up	1,048	High
Hardest photos	screenshots, IDs, bills	106	Med
Who is hit hardest	big libraries, web users	452	High
People photos	faces never processed	260	Med
Did AI search help?	worse for heavy users	520	High

ID	HYPOTHESIS	ENGINE SIGNAL
H1	Remember 2+ cues, type one	79% of searches are 1–2 words
H2	Time is remembered, not usable	scrolling is the top workaround
H3	People photos never processed	89% of those never found
H4	Info photos fail more	70% never found
H5	Context can't be searched	171 stories rely on it
H6	AI search cost heavy users' trust	520 "got worse" posts

Status of all six: **strong signal, test next.** The survey and interviews keep or drop each one.

SEGMENTS · WHY THEY SEARCH

NEVER FOUND

★ Moment seekers

people, pets, trips · 308 stories

77%

Breaks at: faces not processed, misread

Info fetchers

screenshots, bills, IDs · 123 stories

70%

Breaks at: misread; words in image

Video and old scans

clips, printed photos · 94 stories

72%

Breaks at: misread, buried

WHO THEY ARE Big libraries: can't combine cues · Web and desktop: same · iPhone: search got worse

TARGET SEGMENT ★ MOMENT SEEKERS

Largest: biggest of the three groups

Hit hardest: never found more than average

Fixable: they remember who + when, search can't combine them

On focus: exactly the "Describe the memory" problem

> *"I know a very dear picture of my younger sister with our dog in the pool and it didn't find it."*

WHY WE SEGMENTED Same failure, different causes: indexing, misreading, combining cues. **Segments tell us which fix serves whom.**

PART 4 · PRIMARY RESEARCH

Google Photos

Asking people confirmed it: they remember the photo, search can't use what they remember

RECRUITMENT FUNNEL

- **46 surveyed** — all users
- **34 searched recently** — 74%
- **32 qualified** — 70%
- **6 interviews**

Method: survey to screen, then watch a real photo hunt: recall shows in the act

SURVEY · BASE 32 QUALIFIED OF 46

Scrolled the timeline first	69%
Spent 5+ minutes looking	59%
Remembered 2+ details	59%
Found it quickly	22%

Only 1 of 32 typed in search first.

18 of 26 later found looked different from memory.

INTERVIEWS · N=6

- **6/6** went to the timeline or another app first
- **5/6** typed in search; none were helped
- **4/6** gave up on their last lost photo
- **4/6** slowed by look-alike thumbnails
- **2/6** found it in 2 min, only via printed text

Dissent: printed text makes papers easy to find.

H1 · Can't combine cues — Validated

19/32 remembered 2+ details; 5/6 interviewees typed, none helped

Refined: Several details, one search box.

H2 · Time known, not usable — Validated

15/32 remembered rough time, the top cue; 22/32 scrolled back by month

Refined: Time is known but can't be searched.

H3 · People photos not indexed — Refined

2/13 people or event photos found fast; 19/32 use face search

Refined: Faces work; childhood and "beta" don't.

H4 · Info photos fail more — Rejected

0/7 document hunters gave up, though none found it quickly

Refined: Slow, not lost; printed text makes it fast.

H5 · Context can't be searched — Validated

12/32 remembered a life event or source app; neither searchable

Refined: Context is held but unusable.

H6 · AI search costs trust — Too little data

1/46 used Ask Photos; turned it off for exact finds

Refined: Directional only; test in usability.

SYNTHESIS · TRIANGULATED

ENGINE

452 posts: can't combine cues, top area in a near-tie

SURVEY

69% scroll first; only 22% find it fast

INTERVIEWS

4/6 gave up; look-alikes slowed 4/6

Three sources, one gap: people hold several details, search takes one at a time.

Not settled: AI search trust (1 user).

Survey form · Responses · Interview sheet

Note: interviews are hypothetical, built on real survey answers, not recorded calls. Survey is directional, not representative.

PART 5 · PROBLEM DEFINITION

Google Photos

People remember the moment in pieces, and search only takes one piece at a time

1 · TARGET USER SEGMENT

Moment seekers: photos of people, pets, trips, events; India.

Largest group: **308 stories, 77% never found**.
Survey: **5 of 20** found quickly.

2 · PRODUCT OUTCOME TO INFLUENCE

More half-remembered searches end with the photo **opened within 60 seconds**.

Rolls up to the North Star: find rate.

Watch: 2+ detail searches ↑ · rank opened ↓

3 · ROOT CAUSE

Search treats a memory as **one keyword**: it drops the other details and buries the photo among look-alikes, with no reason why.

Not a lost photo: **18 of 26** turned up later. Not a layout gap: albums or sorting won't fix it.

4 · EXISTING WORKAROUNDS

Scroll back month by month: **22 of 32**.

Star, WhatsApp to self, albums: **31 of 32** do at least one.

Interviews: 4 of 6 gave up or asked for a resend.

5 · WHY IT MATTERS TO USERS

19 of 32 spent 5+ minutes; **12** couldn't share it in the moment.

Happens monthly or more for **18 of 32**.

"I knew who, where and when. The app lets me use one at a time."

6 · WHY IT MAKES BUSINESS SENSE

370M searches a month across 1.5B monthly users.

All 4 rivals take a sentence; **none** explains or fixes a match.

Reasoning: unfindable photos weaken the Google One case.

HOW RESEARCH PICKED THE PROBLEM

SEGMENT	BLOCKER	DECISION
Moment seekers · 308	combining details	PURSUED
Info fetchers · 123	unknown wording	NEXT
Video, old scans · 94	indexing (ML)	PARKED

PROBLEM STATEMENT

Moment seekers **want** to find a photo from the few details they recall, **but** search takes one keyword and buries it among look-alikes, **because** it can't combine who, where, when and what's in the photo, or show why a result matched.

PERSONA · COMPOSITE

Aarav, 26
Bengaluru · 9,000 photos

Behaviour: remembers who, where, when; types one word, scrolls

Pain: can't add the second detail

BUSINESS GOAL: Find rate → METRIC TREE: Describe + buried → AI DISCOVERY: 53,986 posts → 10 insights → PRIMARY RESEARCH: 46 → 32 → 6 → PROBLEM: One keyword, many details

Note: persona and interviews are hypothetical, built on real survey respondents. Survey base: 32 qualified of 46.

PART 6 · PRIORITISATION & SOLUTION

Google Photos

Let people describe the photo in their own words, and show why each result matched

WHERE PEOPLE GET STUCK → EIGHT IDEAS → GROUPED INTO THREE CONCEPTS

CAN'T SAY IT ALL	CAN'T SPOT THE RIGHT ONE	GETS NOTHING BACK	CONTEXT CAN'T BE USED
Type the memory in your own words → MEMORY MATCH	Say what each photo matched → MEMORY MATCH	Show near misses if a detail is off → ASK-BACK	Search by a life event → LIFE CHAPTERS
Start from a similar photo → LIFE CHAPTERS	Group look-alike photos → MEMORY MATCH	Ask one follow-up question → ASK-BACK	Search by who sent it → LIFE CHAPTERS

SELECTED · MEMORY MATCH

You type **"me and Riya in Goa around Diwali"**. The app turns it into **tags you can edit**: Riya · Goa · Oct–Nov. It searches with all of them and shows the **6 best photos**, each saying **what matched**: ✓ Riya ✓ Goa × month off. Look-alikes are grouped.

NEXT · FULLER ASK-BACK

A light version is already in the MVP: close matches and one follow-up question.

Next: smarter questions when nothing matches.

NOT NOW · LIFE CHAPTERS

Search by life events ("when we moved house") or by who sent the photo.

Needs new AI and raises privacy questions. Becomes a COULD.

WHAT GOES IN THE MVP

MUST	Your words → editable tags · 6 best photos with what matched · look-alikes grouped · close matches · one follow-up question
ALSO BUILT	Learns your words (photo-checked) · collections · family words · type-ahead
COULD	Sender · similar photo · life events
WON'T	Unscanned photos · backup · replacing Ask Photos

SCORING · THREE CONCEPTS

CONCEPT	VALUE	WORK	SURE	SCORE
Memory Match	5	3	4	**6.7**
Ask-back	4	3	3	4.0
Life Chapters	3	4	3	2.3

Why it wins: it fixes the root cause (search takes one word) using labels Photos already has.

WHY THIS IS NEW

1 · You can see and fix what the app understood. Ask Photos hides it.

2 · Each photo says what matched. No rival app does this.

3 · Still works if you get the date wrong (18 of 26 people did).

4 · AI only reads your sentence; the search uses existing photo labels, so it can't make up results.

Score = (value × how sure we are) ÷ work needed, each rated 1–5 by us from the research

Google Photos

Describe the photo, see why each result came up

MVP OBJECTIVE

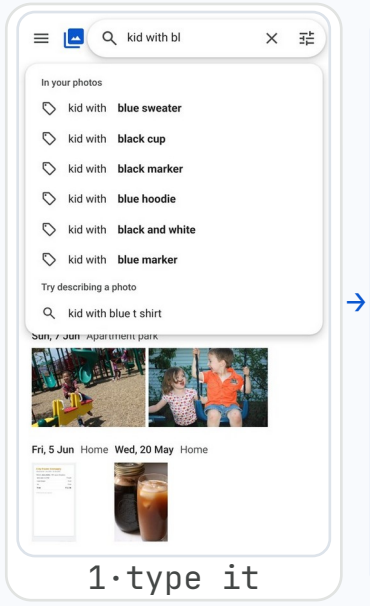
Find a half-remembered photo by **describing it**, and see **why each result came up**.

THE CHANGE

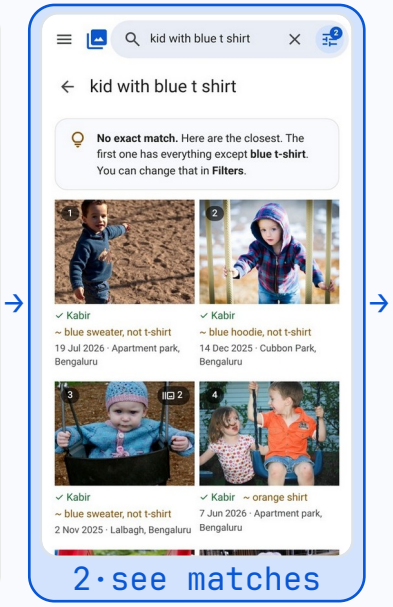
From **one word, endless scrolling** to **the right photo in the top six**.

RULES IT FOLLOWS

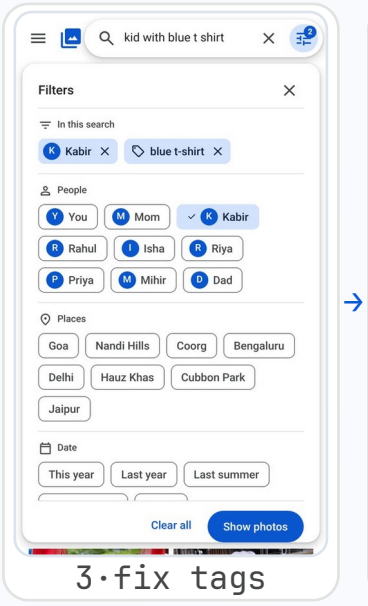
Six best photos, no endless grid
~ close is never shown as exact
Skip words no photo has (“fever”)
Undo any word it learned



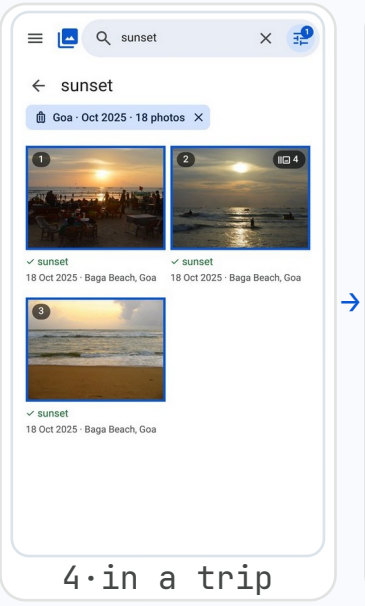
1·type it



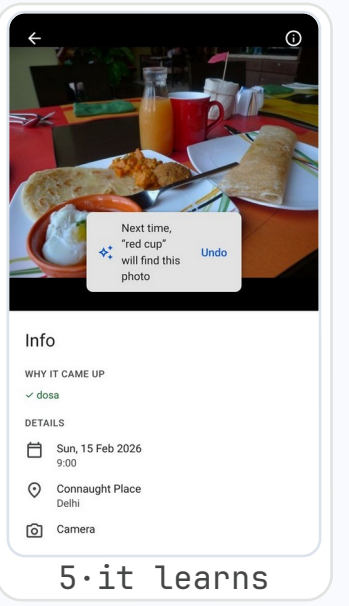
2·see matches



3·fix tags



4·in a trip



5·it learns

FEATURE	WHAT THE USER SEES
Describe it	Type a sentence; Gemini turns it into tags.
Type-ahead	Suggests from your own labels: “dos” → dosa.
What matched	Each photo shows ✓ matched · ~ close · × off.
Close matches	Memory a bit off? A t-shirt still finds a sweater.
Look-alike stacks	Burst shots fold into one tile with a count.
Family words	“kid”, “son”, “beta” mean Kabir; “me” = your shots.

FEATURE	WHAT THE USER SEES
Filters beside ×	Edit tags; add people, place, date, type, words.
One follow-up	Asks who, which year or where, only if needed.
Collections	People, trips, places, things, documents.
Search inside	“goa trip sunset” searches only that trip.
Learns your words	Words are checked against the photo, then saved.
Why it came up	Open a photo: Info lists what matched and why.

UNDER THE HOOD

Search: sentence → Gemini → tags → rules → six best + why.
Learning: missed word → checked → saved to that photo.
Every ✓ is a real label. Any learned word can be undone.

HOW WE CHECKED IT · DEMO NUMBERS, NOT USER RESULTS

6 tasks from the hypothetical interviews: right photo first in 5, second in 1.
27 searches people described in our survey: every one returns the closest photos.

Live prototype · Code on GitHub · 124 openly licensed photos, made-up people, places and dates

Google Photos

Success is more photos found on the first try, not more searches

LEVEL	METRIC	DEFINITION	TARGET
North Star	**Find rate**	Wanted photo opened or shared in 60s ÷ hunts	+15%
Leading	2+ detail searches	Searches with two or more details	60%
Leading	Found in top six	Opened photo was in the first six	≥70%
Leading	Time to open	Seconds from results to opening	-50%
Leading	Searches until found	Searches per photo found	↓
Quality	Full ✓ opens	Opened photo matched every tag	↑
Return	Words kept	Learned words not undone	≥95%
Diagnostic	Close-match finds	Photos found through a ~ match	watch
Guardrail	Wait for results	Slowest 5% of searches	≤2s
Guardrail	Empty searches	Searches that show nothing	not up
Guardrail	Wrong words	Learned words removed as wrong	≤5%
Counter	Follow-up skipped	Follow-up questions dismissed	watch

Events: search_described · tag_edited · results_shown · result_opened · photo_shared · word_learned · word_undone

Targets are proposals, set after a two-week baseline.

KPI TREE · THREE STEPS MULTIPLY INTO FIND RATE

Describe × In top six × Recognised

1,000 hunts × 60% describe × 70% in top six × 85% recognised ≈ **357 found**. A lift early grows every later step.

Full KPI tree with inputs and events per step →

PRE-LAUNCH EVAL GATE

Sentence → tags match a person's	≥90%
Every ✓ traces to a photo label	100%
~ never shown as exact	100%
Learning check agrees with a person	≥95%
Survey searches return closest photos	27/27
Interview tasks: right photo in top two	6/6

27/27 and 6/6 are demo results on the sample library; interviews are hypothetical.

After launch the eval keeps running: each opened photo is checked against its "what matched" line, and misreads are logged for review.

ROADMAP & MOAT

NOW · LIVE MVP
Describe it, what matched, close matches, filters, collections, learns your words.

NEXT
Ask-back: near misses and one question when stuck. Run on real Photos labels, on the phone.

LATER
Life chapters: search by a life event or by who sent it. Voice search.

MOAT
Learned words build a personal memory vocabulary no rival has. Shown reasons build trust.

Experiment: 50/50 holdout · 2-week baseline · 4-week test · stop if any guardrail breaks · KPI tree

Google Photos

Pre-mortem: after launch, people still can't find their photos. Why?

GO-TO-MARKET · NO PAID MEDIA

Phase 1 · beachhead. Moment seekers in India, the largest group. Offer "Describe it" right after a one-word search fails.

Phase 2 · broaden. Everyone who searches. Collections and learned words make the next search faster.

Phase 3 · platform. On the phone and by voice, and from share sheets in chat apps.

WHERE THE VALUE LANDS

USERS	Less time hunting, fewer gave-up searches
PHOTOS	More searches end in a find, so people keep using it
TRUST	Results say why, unlike a black-box answer
LABELS	Learned words show where labels are missing

THE PRODUCT RULE

Never show a guess as a match. Help people find a photo, not search more.

Describe in your words	input
Right photo in top six	mechanism
Find rate	North Star
Wrong words, slow results	guardrails

RISK	MITIGATION	L × I
The AI misreads the sentence	Tags shown and editable in Filters; a rule reader steps in	MED×HIGH BUILT
People don't trust the results	Every photo says what matched; rules, not AI, pick photos	MED×HIGH BUILT
The memory itself is wrong	Close matches (~) and one follow-up question	HIGH×MED BUILT
A wrong word gets learned	Gemini checks the photo first; undo or remove any word	MED×MED BUILT
Sending photos to AI feels unsafe	Check on the phone, ask first, store nothing	MED×HIGH NEXT
Real labels are weaker than the demo's	Run the eval gate on real Photos labels first	HIGH×HIGH NEXT
AI makes search slow or costly	AI reads only the short sentence; 2-second guardrail	MED×MED MONITOR

DE-RISK BEFORE SCALE

1 · PROVE IT HELPS

Today's search vs Memory Match, 50/50. Do more hunts end in a find?

2 · PROVE WHY

≥70% found in the top six, ≥90% of sentences read right, ≤5% wrong words.

3 · SCALE

Roll out market by market, only while every guardrail holds.

It compounds: each find teaches a word → the next search starts closer → more finds.

Final gate: scale it because people **find their photos**, not because they search more.

L × I = likelihood × impact · BUILT = live in the MVP · Limits: made-up labels, hypothetical interviews